

Parametric FWFT synchronous FIFO

1. overview

FIFO (First-in, First-out) is a memory unit used as a buffer between two HW blocks. It is used primarily for two reasons:

- **Rate matching** – when the transmitting block sends data in quick bursts, but the receiver can't process the data fast enough, we add a buffer between them that stores the transmitted data so the receiver can read it in order when it is available. That way the system won't lose data
- **Clock Domain Crossing (CDC)** – Safely passing data between components working with different clocks (Asynchronous FIFO).

This component supports FWFT (First-word Fall-through) to read data with no read latency by storing data in a register near the output so the receiver can read immediately with clock rising instead of waiting another cycle.

The component is fully parameterized for word size and depth flexibility while maintaining timing closure by using separate read/write pointers.

2. parameters

The following parameters are declared during compilation to set the physical structure of the component:

- **DATA_WIDTH** – int > 0: word size in bits (8, 32, 128...) default size is set to 32 bits
- **DEPTH** – int (power of 2): the depth of the FIFO in terms of number of words that can be stored. Default size is set to 16
- **ADDR_WIDTH** – int ($\log_2(\text{DEPTH})$): local parameter

3. Port interface:

3.1. Global signals:

- Clk – input| width=1| main clock.
- rst_n – input| width=1| asynchronous reset. Resets all flags and pointers

3.2. write signals

- wr_en – input| width=1| writing is only possible when wr_en=1 and full=0
- data_in – input| width=DATA_WIDTH| data input for storage
- full – output| width=1| the FIFO is full and can't store more words
- overflow – output| width=1| raising a flag when wr_en=1 and full=1

3.3. read signals

- rd_en – input| width=1| acknowledge reading has happened and can send next word to output reg. only when rd_en=1 and empty=0
- data_out – output| width=DATA_WIDTH| output from storage
- empty – output| width=1| storage is empty and reading is disabled
- underflow – output| width=1| raising a flag when rd_en=1 and empty=1

4. architecture

4.1. memory array – registers array sized [DATA_WIDTH x DEPTH]

4.2. access – supporting writing and reading at the same clock cycle using independent pointers

4.3. pointers and flags – each pointer sized ADDR_WIDTH +1 bits

4.3.1. wr_ptr – MSB represents a full loop was completed. Other bits are address

4.3.2. rd_ptr - MSB represents a full loop was completed. Other bits are address

4.3.3. empty – wr_ptr == rd_ptr (including MSB)

4.3.4. full – wr_ptr = rd_ptr (except MSB)

4.4. FWFT – active pre-fetch. When the output reg is empty/a read was performed, the next word is fetched from the core memory